Jichen Feng

+1 215-452-9136 | jcfeng@sas.upenn.edu | jichen-feng.com | github.com/T3S1AMAX

QUANTITATIVE RESEARCH | NUMERICAL MODELING | SCIENTIFIC COMPUTING

PROFILE

Ph.D. candidate in computational chemistry with a B.S. in physics and experience developing, benchmarking, and validating numerical models for dynamical molecular and solid-state systems. Research spans time-dependent simulation, reduced-model diagnostics, and literature-scale quantitative analysis, including a meta-analysis of CISS literature and research software for mathematical reasoning. First author of multiple peer-reviewed publications in J. Chem. Phys. and Phys. Rev. B, with a strong Python, Julia, C, MATLAB, Linux, and HPC background.

EDUCATION

University of Pennsylvania *Aug. 2021–2027 Spring (expected)*
Ph.D. in Chemistry (Computational Chemistry), GPA: 4.0/4.0 Philadelphia, PA

University of Science and Technology of China *Sept. 2017–Jun. 2021*
B.S. in Physics Hefei, China

QUANTITATIVE AND TECHNICAL SKILLS

Modeling & Analysis: numerical simulation; model benchmarking and validation; time-dependent dynamics; nonlinear response modeling; literature-scale quantitative analysis
Data & Evaluation: structured data extraction; dataset curation and quality control; benchmark design; language-model and agent evaluation; deduplication and tokenization
Programming & Tools: Python, PyTorch, Julia, C, MATLAB, Mathematica, Bash/Shell
Computing & Software: Git, Linux, Slurm, HPC clusters, high-performance and distributed computing; Quantum ESPRESSO, LAMMPS, GAMESS; Microsoft Office, Claude Code, Codex

GRADUATE RESEARCH

- **Small-Molecule Vibrational Chirality Quantification** *May 2023–May 2025*
  Advisors: Prof. Abraham Nitzan, University of Pennsylvania, and Prof. Joseph E. Subotnik, Princeton University
  - Performed first-principle calculation on multiple small molecules with and without chirality.
  - Developed and benchmarked quantitative descriptors of small-molecule vibrational chirality, including continuous chirality measures and momentum-pseudoscalars for normal modes.
- **CISS Transport Modeling** *2024–Present*
  Advisors: Prof. Abraham Nitzan, University of Pennsylvania, and Prof. Joseph E. Subotnik, Princeton University
  - Modeled non-equilibrium vibrational angular momentum and directional energy transfer in driven chiral molecules using analytical models, Langevin dynamics, and classical molecular dynamics.
  - Investigated polaron formation, nonadiabatic electron–nuclear dynamics in chiral molecular systems using mixed quantum–classical models.
  - Performed DFT phonon calculations and mode analysis for chiral CuO material. Study the chiral phonon generation on off-Γ points and predicted the angular momentum generation. Successfully explained the experimental results.
- **Reliability of Ehrenfest Dynamics** *Jul. 2024–Present*
  Advisors: Prof. Joseph E. Subotnik, Princeton University, and Prof. Abraham Nitzan, University of Pennsylvania
  - Benchmarked Ehrenfest dynamics in coupled-oscillator and helical models; showed that enlarging the electronic basis can worsen steady-state populations because ordinary Ehrenfest dynamics does not satisfy quantum detailed balance.
  - Derived and implemented Berry-force corrections for truncated moving electronic bases, restoring linear- and angular-momentum conservation and providing diagnostics for stable reduced models.
- **CISS Literature Meta-Study** *2025–Present*
  Advisors: Prof. Abraham Nitzan, University of Pennsylvania, and Prof. Joseph E. Subotnik, Princeton University
  - Built an AI-assisted extraction, figure-digitization, and verification workflow and curated 6,744 measurements from 279 experimental papers.
  - Designed within-paper and cross-paper statistical analyses across temperature, chiral path length, chiral fraction, magnetic field, voltage, chiroptical strength, measurement method, and material class; identified robust trends and aggregation artifacts.
- **Solid-State Photocurrent Modeling** *May 2022–May 2023*
  Advisor: Prof. Andrew M. Rappe, University of Pennsylvania
  - Developed a real-time photocurrent simulation algorithm for nonlinear optical response in periodic solid-state systems.
  - Applied first-principles electronic-structure calculations to analyze shift-current generation in noncollinear magnetic materials and its dependence on magnetic configuration and symmetry breaking.

UNDERGRADUATE RESEARCH

University of Science and Technology of China *2018–2021*
Undergraduate Researcher – Quantum Information and Condensed Matter Physics Hefei, China

- Designed a state-dependent quantum-cloning scheme and evaluated achievable cloning fidelities; conducted quantum-optics experiments involving PDH locking, magneto-optical traps, single-photon sources, and quantum-state tomography.
- Built density-functional-theory and machine-learning workflows for two-dimensional materials.

INDUSTRY EXPERIENCE

Principia Technologies, Inc. *Jun. 2026–Aug. 2026*
Researcher – Pre-training Data and Model Evaluation San Francisco, CA

- Built a reproducible end-to-end data workflow for mathematical and scientific language-model training, covering heterogeneous ingestion, quality filtering, deduplication, tokenization, distributed processing, and training-ready artifacts.
- Developed and integrated AutoMathText-2.5 and related corpora into model-training experiments and developed evaluation infrastructure for formalized mathematical reasoning and model comparison.

SELECTED AI RESEARCH AND SOFTWARE PROJECTS

Web World Models | *First Author* *2025*

- Designed a simulation framework for persistent, controllable, open-ended agent environments using code-defined state and dynamics with model-generated context and decisions.

Fast Weight Attention for Continual Learning | *Co-author* *2026*

- Trained and evaluated Fast Weight Attention models on algebraic reasoning tasks.

Interactive Benchmarks | *Co-developer* *2026*

- Co-developed an evaluation framework for active information acquisition and budgeted multi-turn reasoning across logic, mathematics, user-interface reconstruction, and strategic environments.

MathCode | *Creator and Lead Developer* *2026–Present*

- Created an open-source verified mathematical-reasoning system combining language models, tool execution, theorem search, Lean 4 verification, and iterative error diagnosis; the project has received 700+ GitHub stars.

PUBLICATIONS AND MANUSCRIPTS

1. Y. Zhang, **J. Feng**, and S. Qin, "Recurrent Looped Transformer," arXiv:2610.07591 (2026).
2. Y. Zhang et al., including **J. Feng**, "Fast Weight Attention for Continual Learning," arXiv:2608.27763 (2026).
3. X. Tao, C. Climent, E. Abraham, **J. Feng**, and A. Nitzan, "Chiral vibrational modes and vibrational circular dichroism," arXiv:2607.22257 (2026).
4. B. Yue et al., including **J. Feng**, "Interactive Benchmarks," arXiv:2603.04737 (2026).
5. **J. Feng**, E. Abraham, J. E. Subotnik, and A. Nitzan, "Vibrational angular momentum generation in thermally driven chiral systems," *Physical Review B* 113, 094301 (2026).
6. **J. Feng**, Y. Zhang, C. Zhang, Y. Lu, S. Liu, and M. Wang, "Web World Models," arXiv:2512.23676 (2025).
7. **J. Feng**, E. Abraham, J. E. Subotnik, and A. Nitzan, "Rectification of vibrational energy transfer in driven chiral molecules," *The Journal of Chemical Physics* 163(23) (2025).
8. **J. Feng**, E. Abraham, J. E. Subotnik, and A. Nitzan, "Chiral vibrational modes in small molecules," *The Journal of Chemical Physics* 163(1) (2025).
9. **J. Feng**, J. E. Subotnik, and A. Nitzan, "On the number of states to include in mixed quantum–classical dynamics: Can more states be worse?" Manuscript in preparation.
10. **J. Feng**, J. E. Subotnik, and A. Nitzan, "An AI-Assisted Meta-Analysis of Chirality-Induced Spin Selectivity." Manuscript in preparation.
11. Aritra Das, **Jichen Feng**, Nazifa Tabassum, Javier Garcia Barriocanal, Abraham Nitzan, Joseph E. Subotnik, Brian Bloom, David H. Waldeck, and Renee Frontiera, "Phonon Angular Momentum Drives Chiral Induced Spin Selectivity," manuscript in preparation.

SELECTED PRESENTATIONS AND AWARDS

CISS Online Seminars, invited research talk	Jun. 2026
APS Global Physics Summit, research talk	Mar. 2026
Gordon Research Conference: Molecular Interactions and Dynamics, poster	Jun. 2026
Gordon Research Conference: Electron Spin Interactions with Chiral Molecules and Materials, poster	Aug. 2025
Graduate Committee's List, University of Pennsylvania	2022
Outstanding Student Scholarship, University of Science and Technology of China	2018–2020